PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Indirect comparison of TIMI, HEART, and GRACE for predicting major cardiovascular events in patients admitted to the emergency department with acute chest pain: A systematic review and meta-analysis
AUTHORS	Ke, Jun; chen, Yiwei; Wang, Xiaoping; Wu, Zhiyong; Chen, Feng

VERSION 1 – REVIEW

REVIEWER	Torralba, Felipe Fundacion Cardioinfantil Instituto de Cardiologia
REVIEW RETURNED	28-Jan-2021

GENERAL COMMENTS	I really liked your work and I congratulate the team because the effort of doing this meta analysis strengthtens the previous findings about this scores, and it hopefully can help physicians all over the world to make decisions in this group of patients, with a higher level of certainty and security, specialy for those patients at the low risk group, in order to optimize resources and save in-hospital time

REVIEWER	Roongsritong, Chanwit Renown Institute for Heart and Vascular Health
REVIEW RETURNED	06-Feb-2021

GENERAL COMMENTS	The author reported the findings of a meta-analysis using pooled data from prospective studies on performances of three commonly used scoring systems (TIMI, HEART and GRACE scores) for acute chest pain risk stratification. The author felt that available data on predictive values of these risk scores are limited and inconclusive and this research would provide needed information. This statement is inaccurate. For the HEART score, multiple previous reports of properly conducted studies (including prospective researches listed in references # 26, 33, 35 and 37 of this manuscript) have clearly illustrated its performance as a very effective risk stratification tool for this common problem among patients presenting to emergency department. Consequently, the HEART score is widely utilized in the United States for this purpose. Furthermore, the GRACE score has been repeatedly shown to have modest predictive value in this situation. Most experts believe as stated by the author that this is likely due to the fact that GRACE score was originally developed to identify high risk individuals among patients presenting with acute coronary syndrome. The finding that TIMI risk score performs as well as HEART score in this report is in contrast to most if not all other major reports

	comparing the two risk scores which showed HEART score to be consistently superior to TIMI risk score. It would be important for the author to elaborate on this discrepancy. As stated by the author, one of the main limitations of pooled analysis is heterogeneity of the data. A 30-day MACE is not clearly defined in this report. Given a large numbers of studies included in this analysis, different definitions of MACE among these studies may have been used. Some if not most of the reports on predictive values of HEART scores (references #26 and 35 for examples) in fact used 6 weeks instead 30 days as a cut-off for MACE. How did the author reconcile the differences if they existed? There are a number of grammatical errors and incorrect statements for examples; The main one is Line 5 in the Introduction section " as CVD is the most common cause of chest pain". This is incorrect and contradicts the next sentence. Line 3; instead of "which can initiate" perhaps more correct to say something like " to ensure timely initiation of proper treatment in..." Line 5 in the Result section; instead of "Moreover" probably should be "Next or Subsequently" etc. Discussion section Line 3; "recruited" should be "included" Line 6; "appropriate" perhaps should be "strong" or "powerful" or "accurate" The author however is to be commended for their effort to advance our knowledge in this very importance topic.

REVIEWER	Berghold, Andrea Medical University of Graz, Institute for Med Inf, Stat and Doc
REVIEW RETURNED	23-Apr-2021

GENERAL COMMENTS	The authors wanted to compare the predictive values of TIMI, HEART, and GRACE for major cardiovascular events in patients admitted to the emergency department with acute chest pain performing a systematic review and meta-analysis. Furthermore, they made an indirect comparison of the three scoring systems. The authors included studies where at least one of the three scoring systems (TIMI, HEART and GRACE) were used. There should be more background regarding the three scoring systems included (i.e. when were they first introduced). The authors should also elaborate on MACE definition. The studies go from 2005 to 2020 although it was stated that the authors included a much longer time-period in their search. This should be explained in more detail. In the first years mainly TIMI studies are found but from 2016 onwards 9 studies included all three scoring systems and the endpoint MACE. It would be of interest to do a subgroup analysis for these studies. The study quality of the included studies was assessed by QUADAS-2 and presented in Table 2. It would be nice to get a graphical display of study quality (cf Fig. 3 of Whiting et al. 2011). Study quality is used for subgroup analysis. It is not clear which studies were classified as high quality or moderate quality. Please

	describe. The section on sensitivity and subgroup analysis should explain more clearly what was done and the motivation for the different subgroups. For the analysis of TIMI 25 studies were included. Looking at the Fig. S1 there seems to be studies with very homogeneous results (study 1-13) while study 14-25 show more variability. One cannot identify the studies and the same is true for the other two figures S2 and S3 and Supplement 2. In the discussion section the direct comparison of the three scoring systems should be elaborated in more detail. Heterogeneity is only partly addressed. Page 7 Indirect comparison: replace Indirectindirect by Indirect

VERSION 1 – AUTHOR RESPONSE

Response to Reviewer #1

General comments: I really liked your work and I congratulate the team because the effort of doing this meta analysis strengthens the previous findings about this scores, and it hopefully can help physicians all over the world to make decisions in this group of patients, with a higher level of certainty and security, specialy for those patients at the low risk group, in order to optimize resources and save in-hospital time

Response: We appreciate the reviewer given this kindly comment.

Response to Reviewer #2

General comments: The author reported the findings of a meta-analysis using pooled data from prospective studies on performances of three commonly used scoring systems (TIMI, HEART and GRACE scores) for acute chest pain risk stratification. The author felt that available data on predictive values of these risk scores are limited and inconclusive and this research would provide needed information. This statement is inaccurate.

Response: As behalf of all co-authors, I would like to appreciate this referee due to thoughtful comments proposed by the peer review. After we revised the manuscript, those significant issues could be changed. Moreover, the statement for predictive values of these risk scores are limited and inconclusive have already changed into: “However, the predictive values of risk stratification measured by the TIMI, HEART, and GRACE scoring systems on MACEs have not been fully compared.”

Question 1: For the HEART score, multiple previous reports of properly conducted studies (including prospective researches listed in references # 26, 33, 35 and 37 of this manuscript) have clearly illustrated its performance as a very effective risk stratification tool for this common problem among patients presenting to emergency department. Consequently, the HEART score is widely utilized in the United States for this purpose.

Furthermore, the GRACE score has been repeatedly shown to have modest predictive value in this situation. Most experts believe as stated by the author that this is likely due to the fact that GRACE score was originally developed to identify high risk individuals among patients presenting with acute coronary syndrome.

The finding that TIMI risk score performs as well as HEART score in this report is in contrast to most if not all other major reports comparing the two risk scores which showed HEART score to be consistently superior to TIMI risk score.
It would be important for the author to elaborate on this discrepancy.
Response: Thanks for this suggestion. Several sentences have already added in the Discussion section to elaborate the discrepancy among TIMI, GRACE, and HEART.
Question 2: As stated by the author, one of the main limitations of pooled analysis is heterogeneity of the data. A 30-day MACE is not clearly defined in this report. Given a large numbers of studies included in this analysis, different definitions of MACE among these studies may have been used. Some if not most of the reports on predictive values of HEART scores (references #26 and 35 for examples) in fact used 6 weeks instead 30 days as a cut-off for MACE. How did the author reconcile the differences if they existed?
Response: Thanks for this suggestion. We know the definition of MACEs and follow-up duration are important sources for the heterogeneity among included studies. The definition of MACEs has already been presented in Table 1. Moreover, subgroup analysis based on follow-up duration have already performed, and the results are shown in Table 4. The 30 days of follow-up was defined as cutoff-value owing to it was commonly used for assessing the MACEs occurred in hospital. We have already addressed this comment in Limitation section.
Question 3: There are a number of grammatical errors and incorrect statements for examples;
The main one is Line 5 in the Introduction section " as CVD is the most common cause of chest pain". This is incorrect and contradicts the next sentence.
Line 3; instead of "which can initiate" perhaps more correct to say something like " to ensure timely initiation of proper treatment in..."
Line 5 in the Result section; instead of "Moreover" probably should be "Next or Subsequently" etc.
Discussion section: Line 3; "recruited" should be "included"
Line 6; "appropriate" perhaps should be "strong" or "powerful" or "accurate"
Response: Thanks for this suggestion. We have already made these changes in the revised manuscript. Moreover, the English revision have already performed in Editage Company.
Question 4: The author however is to be commended for their effort to advance our knowledge in this very importance topic.
Response: We appreciate the reviewer given this kindly comment.

Response to Reviewer #3

General comments: The authors wanted to compare the predictive values of TIMI, HEART, and GRACE for major cardiovascular events in patients admitted to the emergency department with acute chest pain performing a systematic review and meta-analysis. Furthermore, they made an indirect comparison of the three scoring systems.
Response: As behalf of all co-authors, I would like to appreciate this referee due to thoughtful comments proposed by the peer review. After we revised the manuscript, those significant issues could be changed.
Question 1: The authors included studies where at least one of the three scoring systems (TIMI, HEART and GRACE) were used. There should be more background regarding the three scoring systems included (i.e. when were they first introduced).
Response: Thanks for this suggestion, and the background for three scoring systems have already added in Introduction section.
Question 2: The authors should also elaborate on MACE definition.

Response: Thanks for this suggestion, and the details regarding the MACE definition in individual trial are shown in Table 1.

Question 3: The studies go from 2005 to 2020 although it was stated that the authors included a much longer time-period in their search. This should be explained in more detail.

Response: Thanks for this suggestion, and the following sentences have already added in the revised manuscript: “Risk stratification by the TIMI score was available in 25 studies and published between 2005 to 2020, 16 studies used the HEART score and published between 2013 to 2020, and 16 studies employed GRACE and published between 2007 to 2020”

Question 4: In the first years mainly TIMI studies are found but from 2016 onwards 9 studies included all three scoring systems and the endpoint MACE. It would be of interest to do a subgroup analysis for these studies.

Response: Thanks for this suggestion. Sensitivity analysis were performed for studies directly comparisons 3 scoring systems and the results shown in Table 3.

Question 5: The study quality of the included studies was assessed by QUADAS-2 and presented in Table 2. It would be nice to get a graphical display of study quality (cf Fig. 3 of Whiting et al. 2011). Study quality is used for subgroup analysis. It is not clear which studies were classified as high quality or moderate quality. Please describe.

Response: Thanks for this suggestion, and the study quality of the included studies have already presented in graphical. Moreover, individual study with 3 items with high risk or unclear were considered as moderate quality.

Question 6: The section on sensitivity and subgroup analysis should explain more clearly what was done and the motivation for the different subgroups.

Response: Thanks for this suggestion, and the results of sensitivity and subgroup analysis have already changed in the revised manuscript.

Question 7: For the analysis of TIMI 25 studies were included. Looking at the Fig. S1 there seems to be studies with very homogeneous results (study 1-13) while study 14-25 show more variability. One cannot identify the studies and the same is true for the other two figures S2 and S3 and Supplement 2.

Response: Thanks for this suggestion. We have already replaced the number to author’s name and publication year in forest plots, and these results might due to chance.

Question 8: In the discussion section the direct comparison of the three scoring systems should be elaborated in more detail.

Response: Thanks for this suggestion. The results for direct comparison the 3 scoring systems have already calculated in Table 3. Moreover, several sentences have already added in Discussion section.

Question 9: Heterogeneity is only partly addressed.

Response: Thanks for this suggestion, and the following sentence have already added in Limitation section: “Second, substantial heterogeneity was detected across the included studies, and the heterogeneity were not fully explained by sensitivity and subgroup analyses, which might be attributed to the various definitions of MACEs, follow-up durations, and individual risk level for MACEs”.

Question 10: Page 7 Indirect comparison: replace Indirectindirect by Indirect

Response: Thanks for this suggestion. We have already made this change in the revised manuscript.

VERSION 2 – REVIEW

REVIEWER	Roongsritong, Chanwit

	Renown Institute for Heart and Vascular Health
REVIEW RETURNED	04-Jun-2021

GENERAL COMMENTS	In response to prior review, the authors have made several notable changes in the manuscripts. Overall it is reasonably well written although the followings should be addressed; 1. The discrepancy in the performance of TIMI and HEART score when all studies were included vs only when those using all three scores were analyzed should be more elaborated. The results of most previous direct comparison studies showed HEART score to be superior to TIMI score. The statement that results of subgroup analyses were consistent with overall results is perhaps partially inaccurate. 2. The discussion section should be more concise regarding to the results of previous studies. 3. Last sentence of the first paragraph in discussion section "Finally, the prevalence of MACEs lower than expected" was unclear on which study the authors are referring to. 4. First paragraph on page 12 in the discussion section, it is unclear why the authors switched from HEART to GRACE. Their discussion here also are somewhat incoherent. 5. Higher odds ratio for HEART than TIMI despite similar AUC perhaps should be briefly elaborated. 6. It is probably not necessary to spell out the criteria well-known TIMI and HEART scores on page 1 in the introduction.

REVIEWER	Berghold, Andrea Medical University of Graz, Institute for Med Inf, Stat and Doc
REVIEW RETURNED	01-Jun-2021

GENERAL COMMENTS	The authors responded well to the Reviewer comments. Regarding MACE the authors have added in the inclusion criteria that a clear definition of MACEs had to be provided in the study. The authors included a wide range of definitions (see Table 1) and different follow-up time which was only elaborated on very shortly in the discussion section. Maybe some more discussion on this part is needed. Furthermore, there are only some minor issues remaining: Figure 2 is missing. In the paragraph Study characteristics there is the word ”were” missing twice in the new sentence. In the discussion section please change “have relatively powerful predictive values”.

VERSION 2 – AUTHOR RESPONSE

Response to Reviewer #3

General comments: The authors responded well to the Reviewer comments. Regarding MACE the authors have added in the inclusion criteria that a clear definition of MACEs had to be provided in the study. The authors included a wide range of definitions (see Table 1) and different follow-up time which was only elaborated on very shortly in the discussion section. Maybe some more discussion on this part is needed.

Response: Thanks for this suggestion, we known MACE definition and follow-up durations could affect the predictive value of TIMI, HEART, and GRACE. Subgroup analysis based on follow-up durations have already performed, while the predictive value according to MACEs definition was not conducted owing to varies outcome definitions across included studies. Furthermore, several sentences have already added in Discussion section to address the potential impact of MACEs definition and follow-up durations.

Question 1: Furthermore, there are only some minor issues remaining:

Figure 2 is missing. In the paragraph Study characteristics there is the word ”were” missing twice in the new sentence.

In the discussion section please change “have relatively powerful predictive values”.

Response: Thanks for this suggestion, and these changes have already performed in the revised manuscript.

Response to Reviewer #2

General comments: In response to prior review, the authors have made several notable changes in the manuscripts. Overall it is reasonably well written although the followings should be addressed;

Response: As behalf of all co-authors, I would like to appreciate this referee due to thoughtful comments proposed by the peer review.

Question 1: The discrepancy in the performance of TIMI and HEART score when all studies were included vs only when those using all three scores were analyzed should be more elaborated. The results of most previous direct comparison studies showed HEART score to be superior to TIMI score. The statement that results of subgroup analyses were consistent with overall results is perhaps partially inaccurate.

Response: Thanks for this suggestion. We have already added several sentences in Discussion section to address the results of sensitivity and subgroup analysis.

Question 2: The discussion section should be more concise regarding to the results of previous studies.

Response: Thanks for this suggestion, and several sentences regarding the results of previous studies have already removed in the revised manuscript.

Question 3: Last sentence of the first paragraph in discussion section "Finally, the prevalence of MACEs lower than expected" was unclear on which study the authors are referring to.

Response: Thanks for this suggestion, and this sentence have already changed into: "Finally, the prevalence of MACEs during follow-up in these studies was lower than expected, which caused broad 95% CIs; i.e., no statistically significant difference [26,35,36,45]."

Question 4: First paragraph on page 12 in the discussion section, it is unclear why the authors switched from HEART to GRACE. Their discussion here also are somewhat incoherent.

Response: Thanks for this suggestion, and this paragraph have already divided into 2 paragraphs, and several sentences have already changed.

Question 5: Higher odds ratio for HEART than TIMI despite similar AUC perhaps should be briefly elaborated.

Response: Thanks for this suggestion, and the predictive values of HEART and TIMI have already illustrated in Discussion section.

Question 6: It is probably not necessary to spell out the criteria well-known TIMI and HEART scores on page 1 in the introduction.

Response: Thanks for this suggestion, and several sentences have already deleted in Introduction section.

VERSION 3 – REVIEW

REVIEWER	Roongsritong, Chanwit Renown Institute for Heart and Vascular Health
REVIEW RETURNED	26-Jun-2021

GENERAL COMMENTS	Congratulations.

REVIEWER	Berghold, Andrea Medical University of Graz, Institute for Med Inf, Stat and Doc
REVIEW RETURNED	23-Jun-2021

GENERAL COMMENTS	Change America to American.

VERSION 3 – AUTHOR RESPONSE

Response to Reviewer #3

General comments: Change America to American

Response: Thanks for this suggestion. We have already made this change in the revised manuscript.

Response to Reviewer #2

General comments: Congratulations.

Response: We appreciate the reviewer given this kindly comment.